

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Applied AI), EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Designed production generative AI systems using LLMs and retrieval-augmented generation (RAG) to power enterprise user workflows, processing 15M+ interaction records improving response relevance by 30%.
- Developed multimodal AI pipelines combining transformer-based NLP and vision models to enable semantic search, document understanding, and cross-modal retrieval in production AI services.
- Implemented retrieval-augmented generation (RAG) pipelines integrating LLMs (GPT-4, LLaMA, Gemini) with vector search and contextual memory to improve reasoning quality and response accuracy by 30%.
- Built evaluation and monitoring frameworks using experiment tracking, automated regression testing, and production telemetry to continuously measure model quality, latency, and safety.
- Built and operated scalable ML services using FastAPI and cloud infrastructure, enabling model training, evaluation, and real-time inference with CI/CD workflows and automated monitoring.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data pipelines for feature engineering, dataset validation, and model training, enabling repeatable experimentation and reliable deployment workflows.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, and indexed lookups to efficiently clean, deduplicate, transform large datasets, reducing 50% production issues.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning: End-to-end ML pipelines, model training, evaluation, monitoring, feature engineering, MLOps.
GenAI & LLMs: RAG, prompt engineering, LLM evaluation, context orchestration, multimodal AI systems.
AI Systems: Vector search, semantic retrieval, experiment design, model monitoring, production ML pipelines.
Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Pinecone, Chroma.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Multimodal Retrieval and Question Answering System (Production): Built LLM-powered retrieval pipelines combining embeddings, vector search, and multimodal inputs to deliver grounded responses in production AI workflows.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>